Geospatial distributions reflect rates of evolution of features of language

Henri Kauhanen[1], Deepthi Gopal[2], Tobias Galla[3] & Ricardo Bermúdez-Otero[1]

[1]School of Arts, Languages and Cultures, University of Manchester
[2]Department of Theoretical and Applied Linguistics, Cambridge University
[3]School of Physics and Astronomy, University of Manchester

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This talk

- Question: How can the rate of change (= cultural evolution rate) of a linguistic feature be estimated?
- Previous techniques based on a phylogenetic approach with serious limitations
- Let's try something new:
  - Formulate a model of language change that incorporates spatial dynamics
  - Apply model to available spatial data (from WALS) to infer evolution rates
  - Cross-check against previous approaches and existing literature
- Heuristic: unstable features are spatially scattered, whereas stable ones are found in clusters

Previous methods

- Most existing methods for evolution rate estimation based on phylogenetics
- Heuristic: stable features are conserved within language families. For unstable features, there is within-family variation
- Different implementations of this basic idea,[1] all
  1. requiring a prior genetic family classification, and
  2. neglecting spatial interactions between contiguous speech communities

[1] E.g. Maslova (2004), Wichmann and Holman (2009), Greenhill, Atkinson, Meade, and Gray (2010), Dediu (2011), Greenhill et al. (2017); see Dediu and Cysouw (2013) for a useful overview

Example: Wichmann & Holman's Metric C

> *if one given feature more often tends to have the same value for languages that are related than does another given feature, then the first of the two may be considered to be more stable*[2]

- For genus G, compute proportion of language pairs sharing the feature, R_G
- Average across genera, arriving at R
- Compute proportion of agreeing pairs across **un**related languages, U
- Feature's stability is given by

$$S = \frac{R - U}{1 - U}$$

[2] Wichmann and Holman (2009, p. 16)

Typical cases

- In a metastudy by Dediu and Cysouw (2013), the following (inter alia) are stable:
  - order of object and verb
  - order of subject and verb
  - front rounded vowels
  - tone
- and the following unstable:
  - articles
  - number of cases
  - perfect tense
  - size of consonant inventory

The phylogenetic problem

- Not so much about the reliability of family classifications
- Rather that it's very difficult to ascertain that the families used are of similar time depths

> *The notion genus is explained in Dryer (1989). It is intended as a level of classification which is comparable across the world [...] The decisions as to which groups to treat as genera here are best described as my own educated guesses. [...] in the absence of a tradition within the field of attempting to identify groups of comparable time depth in different parts of the world, they should not be considered more than educated guesses.*[3]

[3] https://wals.info/languoid/genealogy

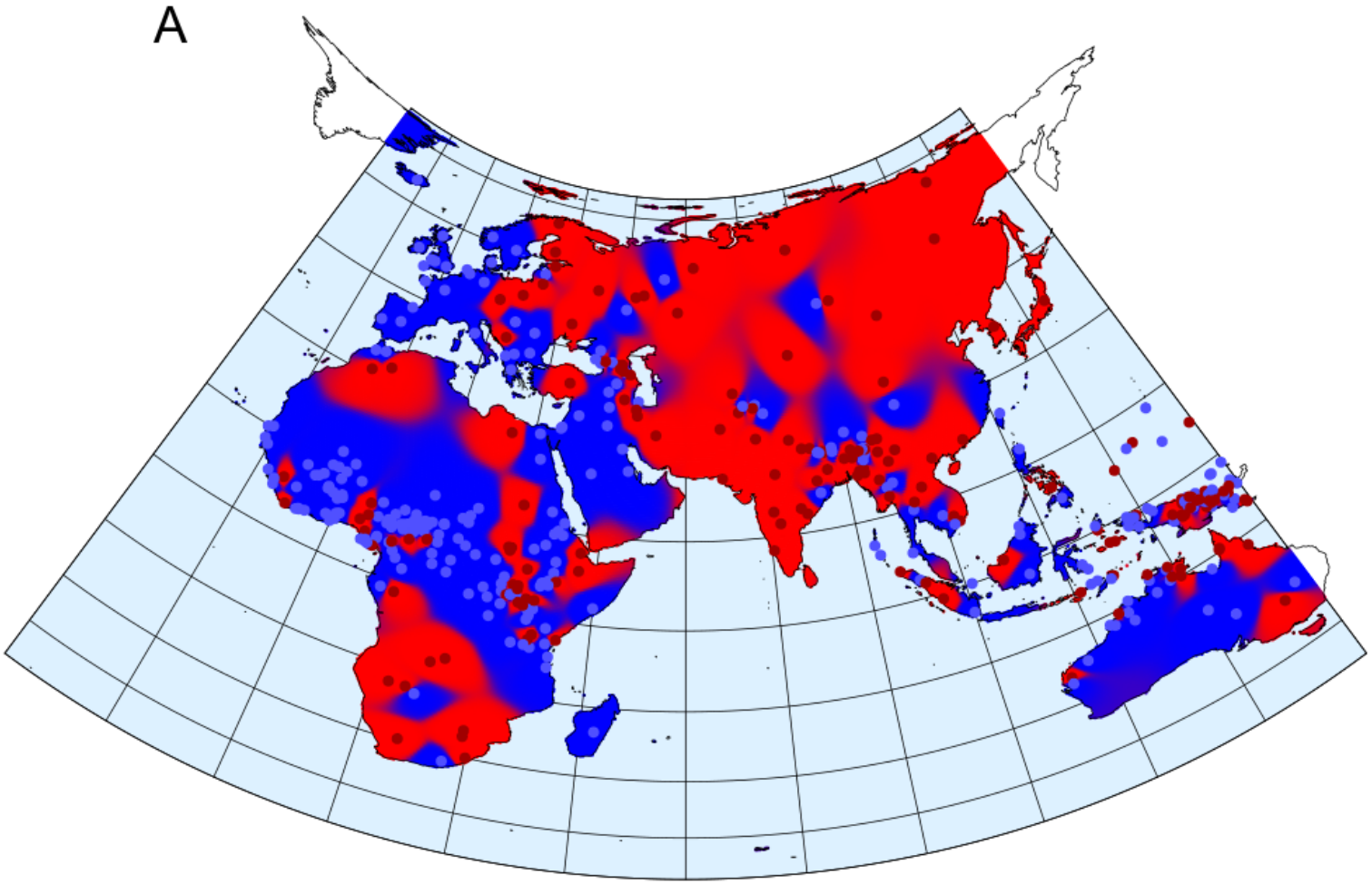
The spatial problem

- Not so much that spatial interactions exist
- Rather that some features are more prone to them than others
  - Case systems are vulnerable to simplification in contact situations involving L2 learners (Bentz & Winter, 2013)
  - Inflectional markers are resistant to horizontal transfer (Gardani, Arkadiev, & Amiridze, 2014)

Proposal

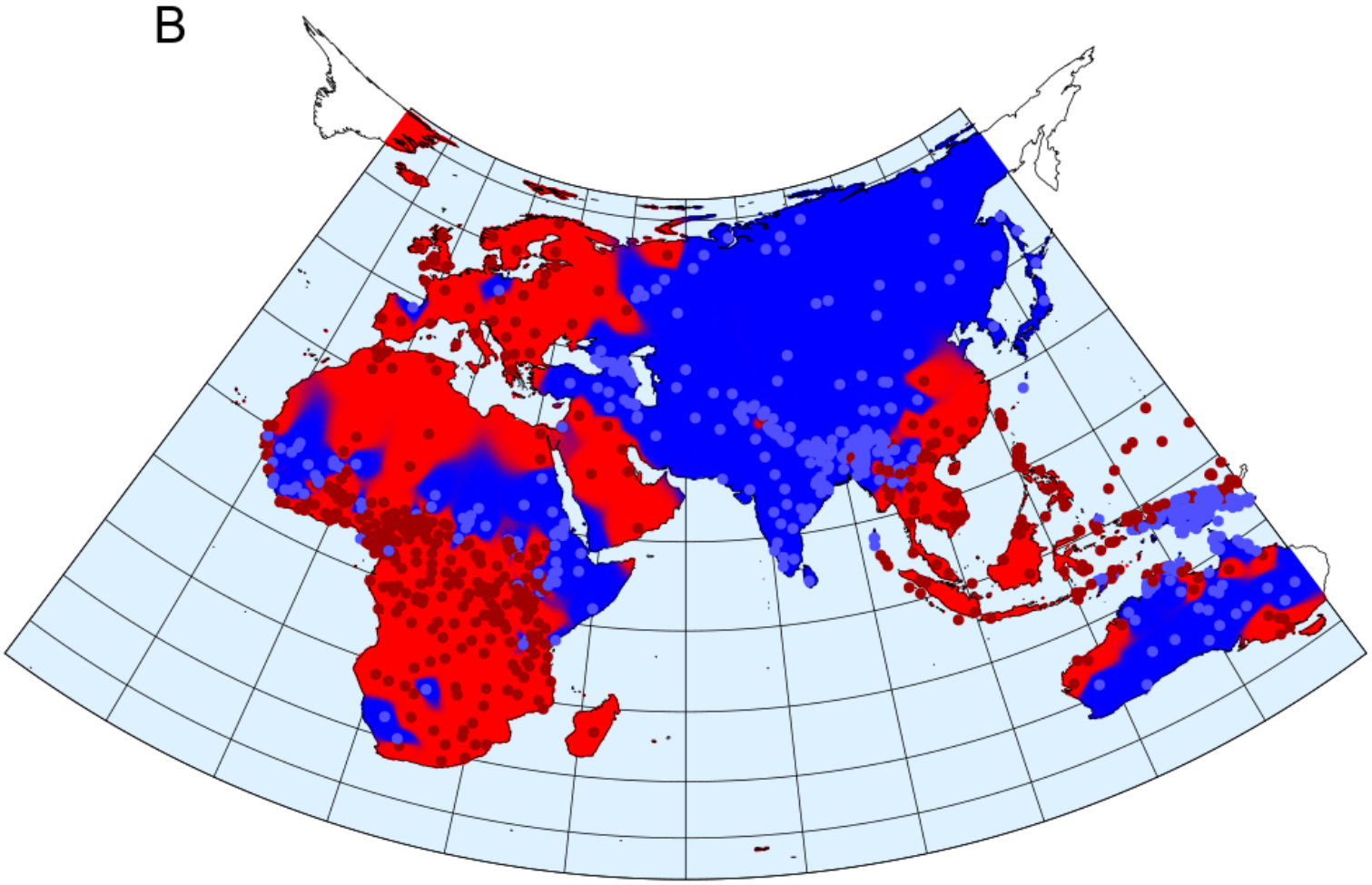
Perhaps educated guesses about phylogenies can be traded for a reliable spatial signal?

definite article (WALS 37A)

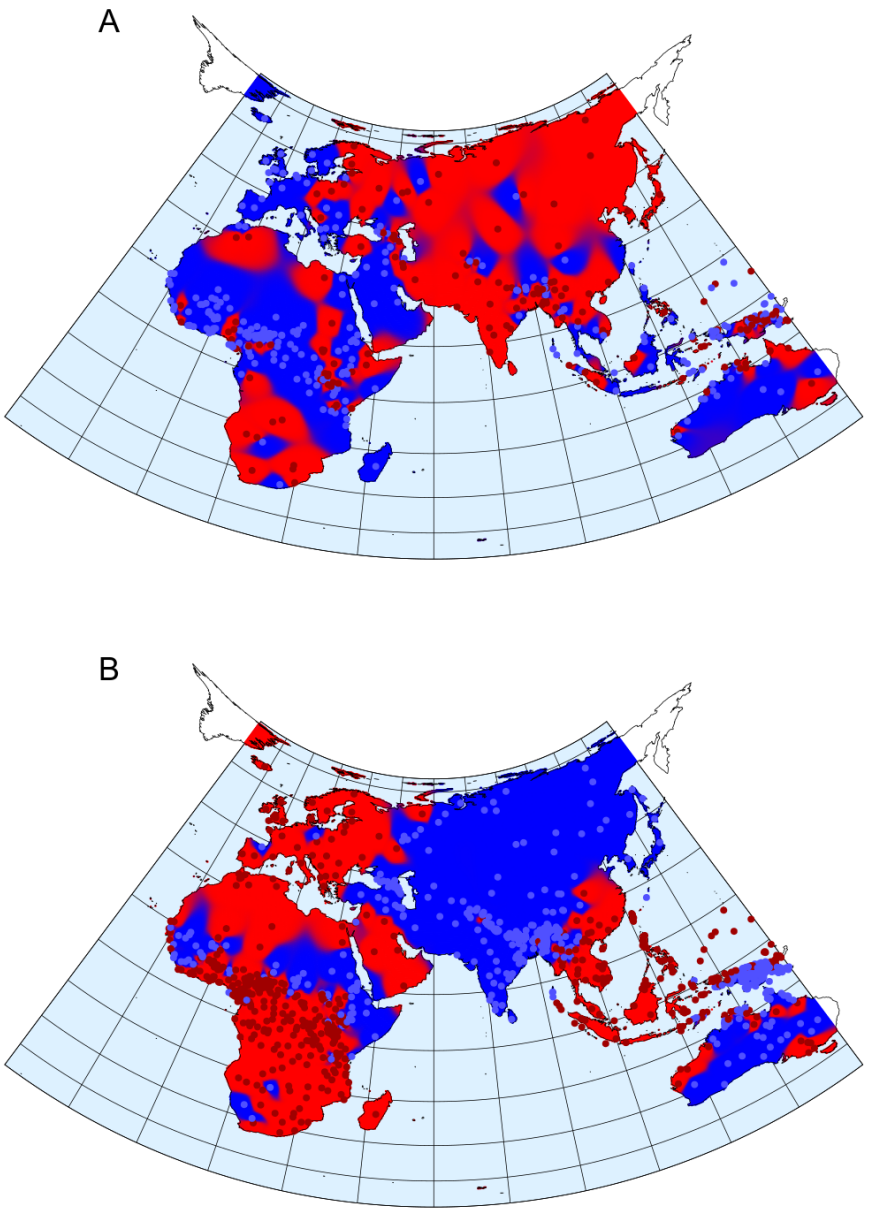


data from Dryer and Haspelmath (2013),
interpolated equal-area Albers projection

OV order (WALS 83A)

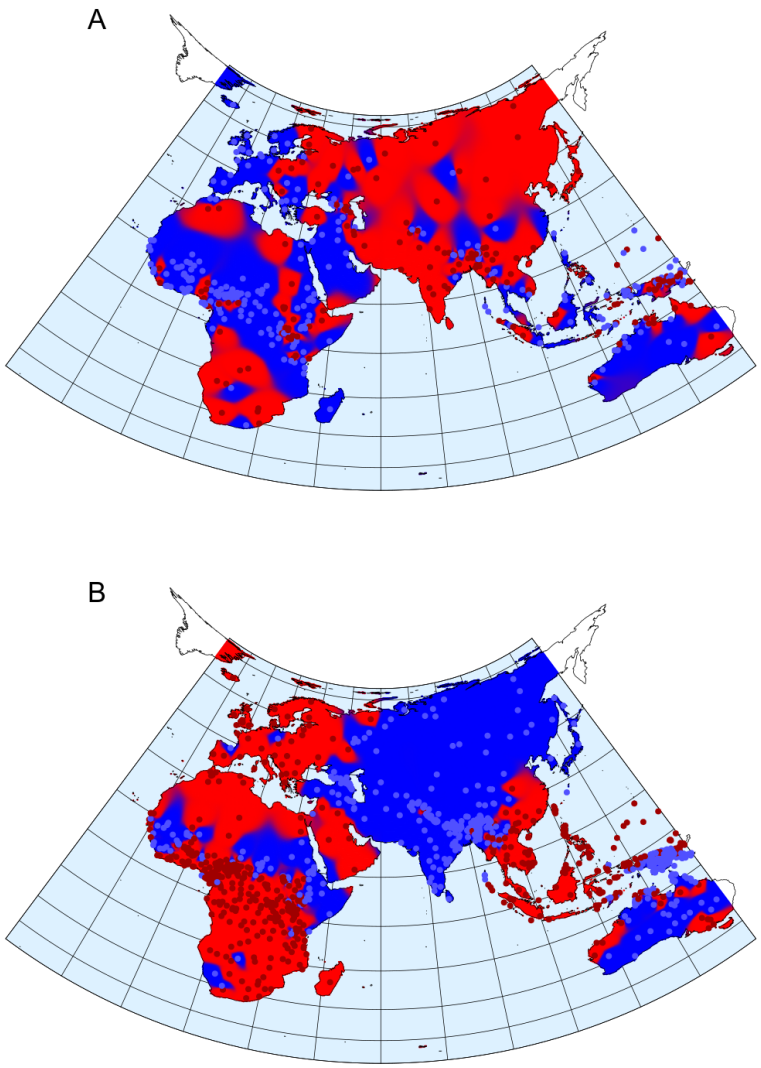


data from Dryer and Haspelmath (2013),
interpolated equal-area Albers projection



Heuristic:
Unstable features scatter, stable features cluster

A measure of spatial relations



- How to quantify this?
- **Isogloss density** σ:

$$\sigma = \text{Prob(two neighbouring languages disagree)}$$

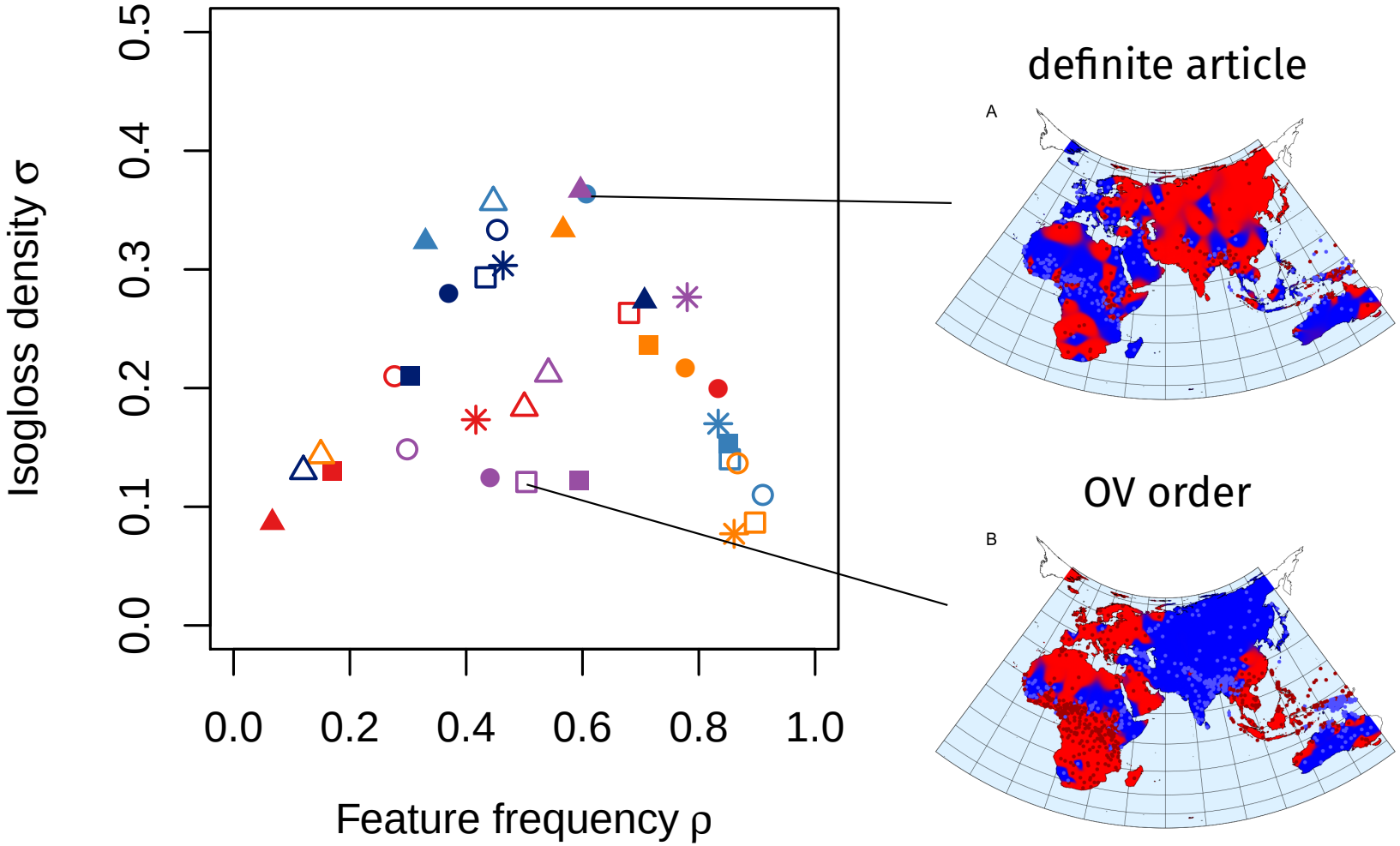
- Neighbours determined by great-circle distance from WALS coordinate data
- We will also record the overall **feature frequency** ρ

Our features

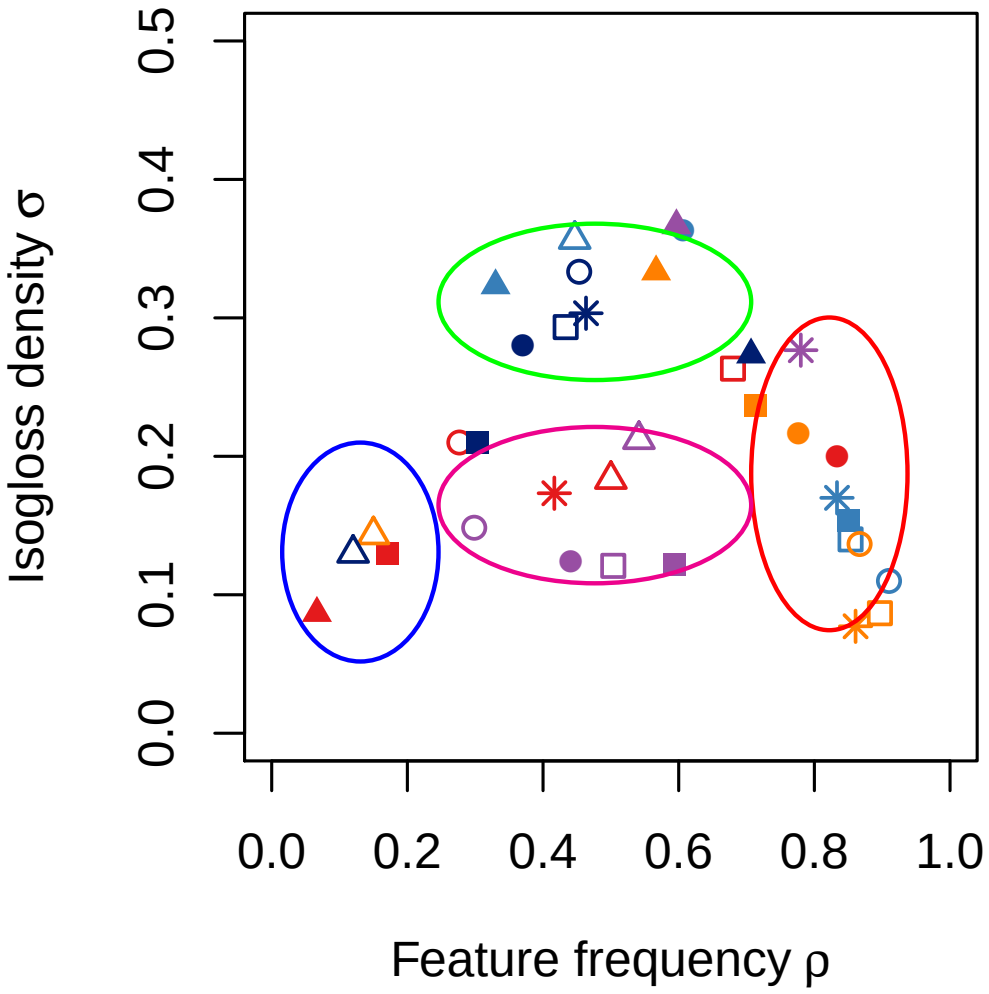
- 35 binary (or binarized) features from WALS, recorded for $\geq$ 300 languages across the world:

1. voicing contrast
2. uvular consonants
3. glottalized consonants
4. lateral consonants
5. velar nasal
6. front rounded vowels
7. tone
8. inflectional morphology
9. productive reduplication
10. plural
11. definite article
12. indefinite article
13. gender distinctions in independent personal pronouns
14. adpositions
15. ordinal numerals
16. possessive affixes
17. tense–aspect inflection
18. morphological second-person imperative
19. inflectional optative
20. grammatical evidentials
21. question particle
22. verbal person marking
23. order of subject and verb is SV
24. order of object and verb is OV
25. order of genitive and noun is GenN
26. order of adjective and noun is AdjN
27. order of numeral and noun is NumN
28. order of degree word and adjective is DegAdj
29. preverbal negative morpheme
30. postverbal negative morpheme
31. passive construction
32. shared encoding of nominal and locational predication
33. zero copula for predicate nominals
34. *hand* and *arm* identical
35. *hand* and *finger(s)* identical

The WALS data (35 features)



Greenberg's (1978) Dynamic Typology

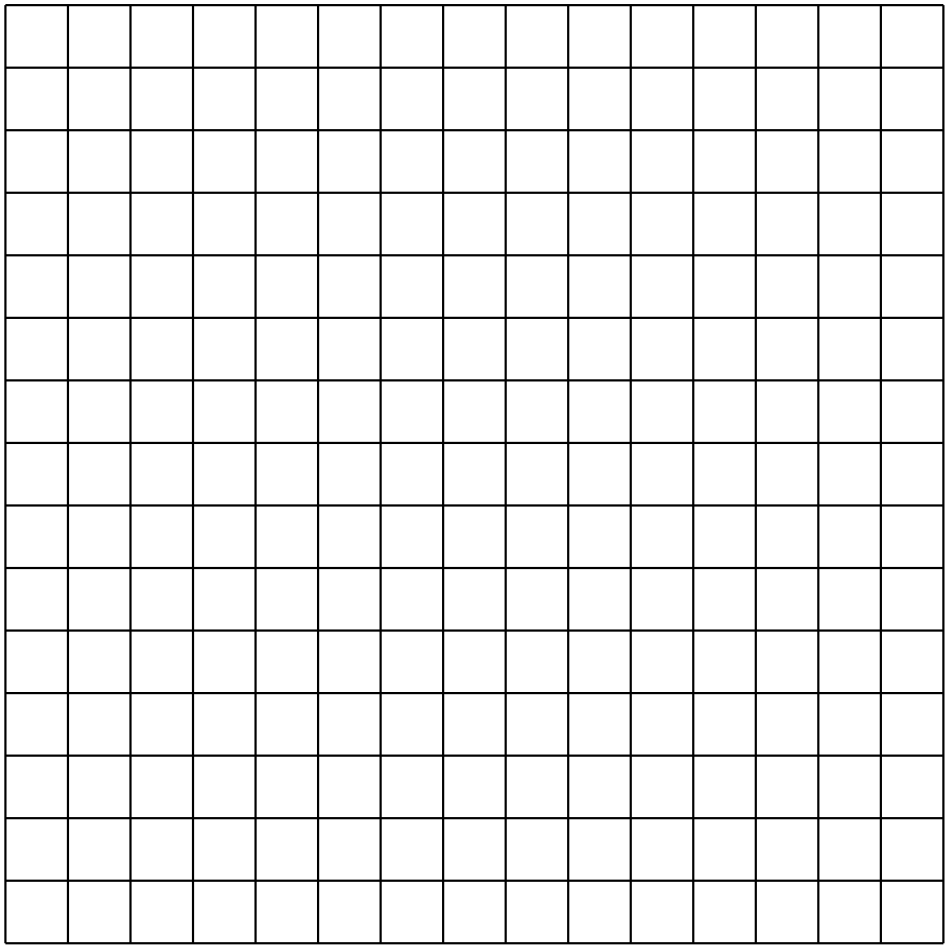


	ingress	egress
I	high	low
II	low	high
III	high	high
IV	low	low

Explicating Greenberg

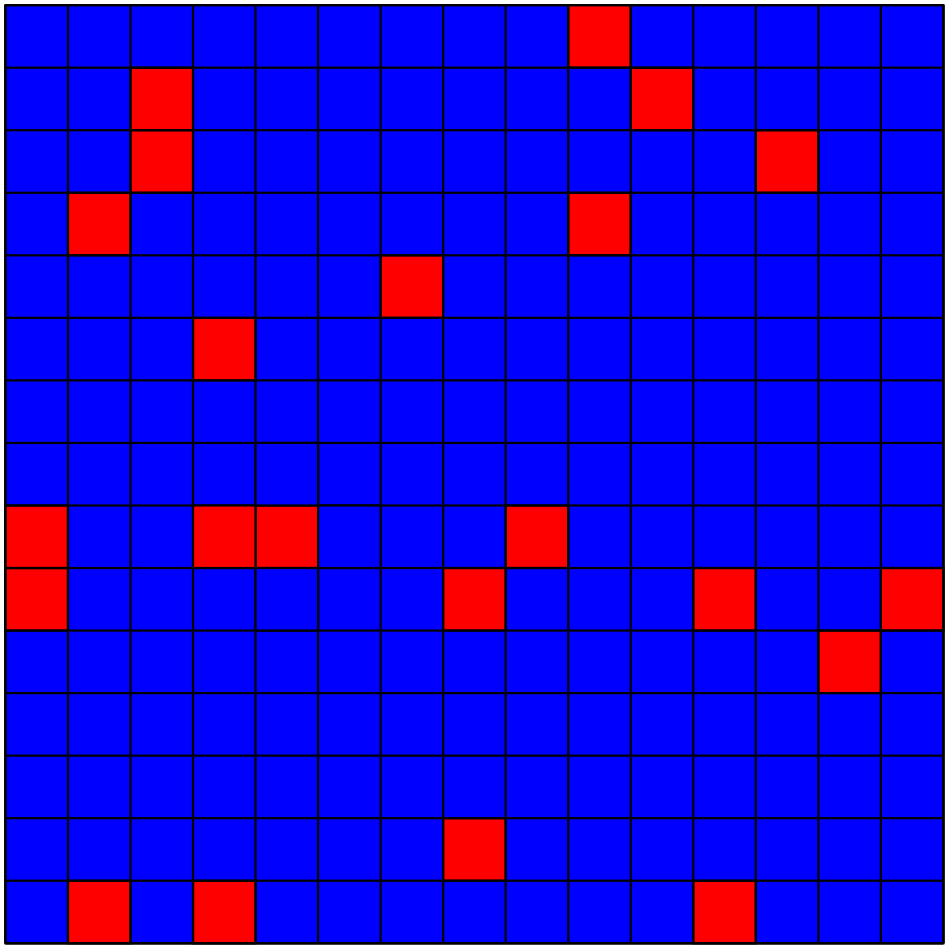
- Greenberg's proposal does not mention any spatial dynamics
- Let's formalize Greenberg, adding a very simple spatial process to the picture
- Model combines:
  - Stochastic "Greenbergian" process for ingress and egress
  - Classical "voter model" for spatial interactions (see Castellano, Fortunato, & Loreto, 2009, for an overview)

Model



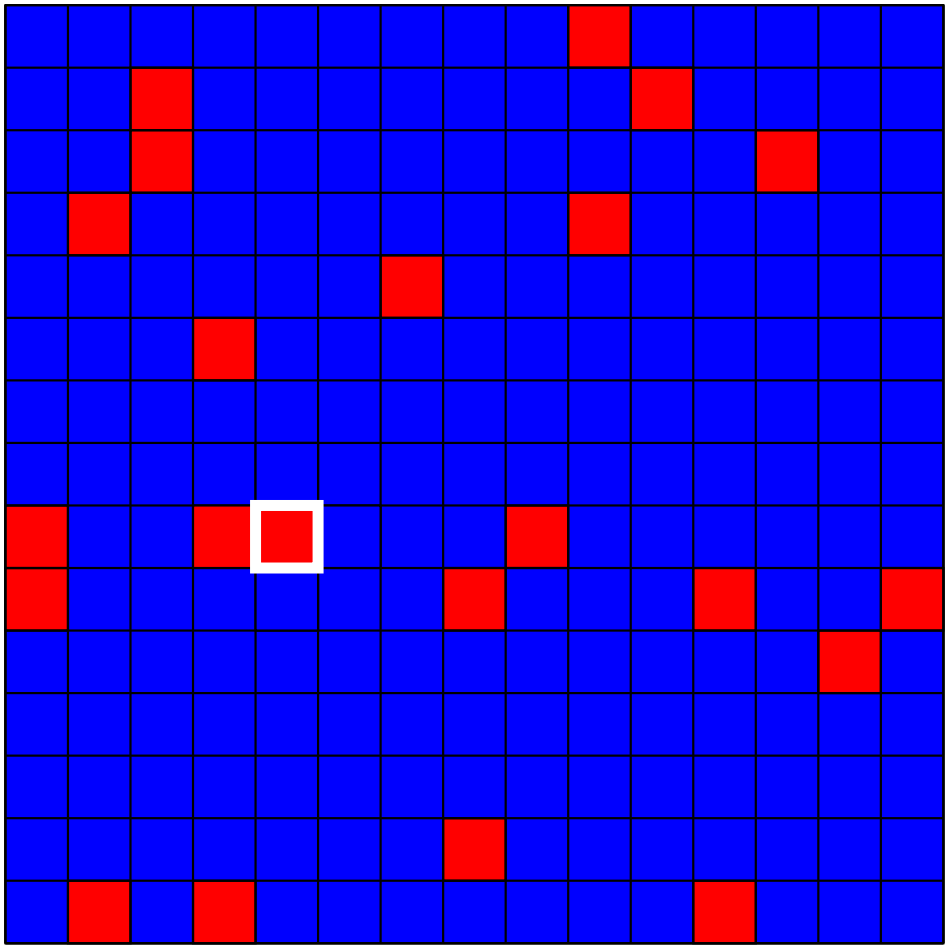
regular lattice with periodic boundary conditions

Model



regular lattice with periodic boundary conditions

Model

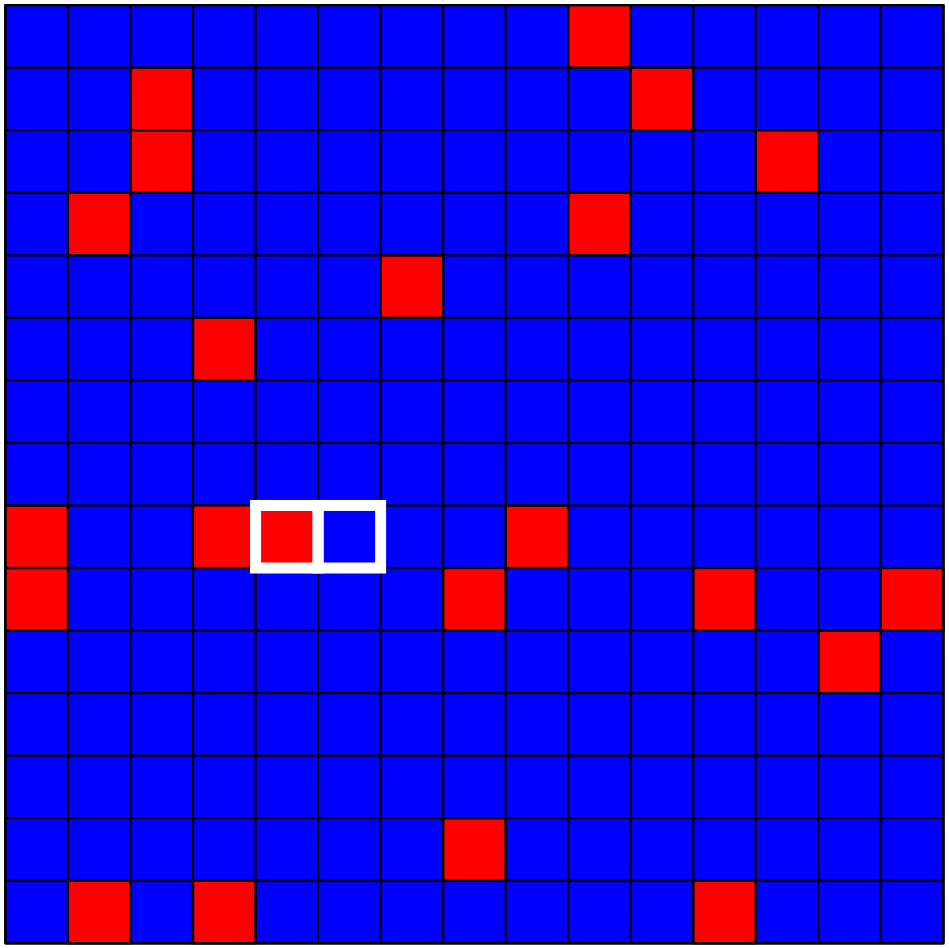


regular lattice with periodic boundary conditions

with prob. q:

pick random site

Model



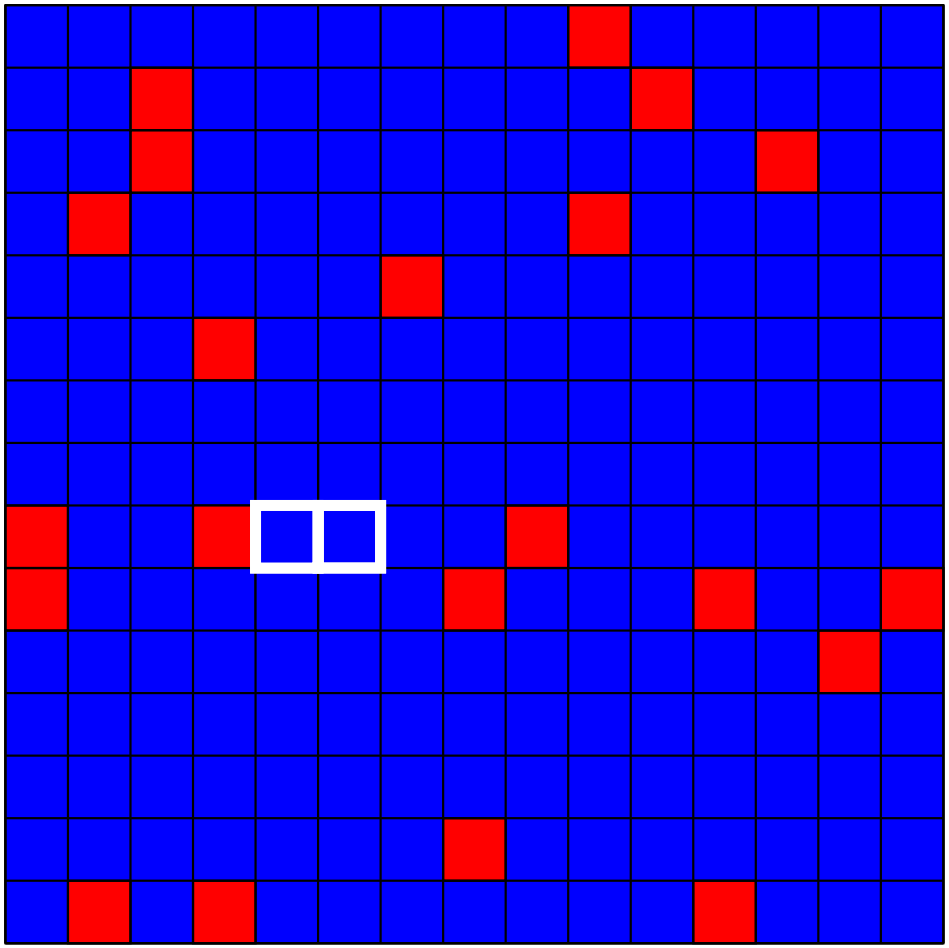
regular lattice with periodic boundary conditions

with prob. q:

pick random site

pick random neighbour

Model

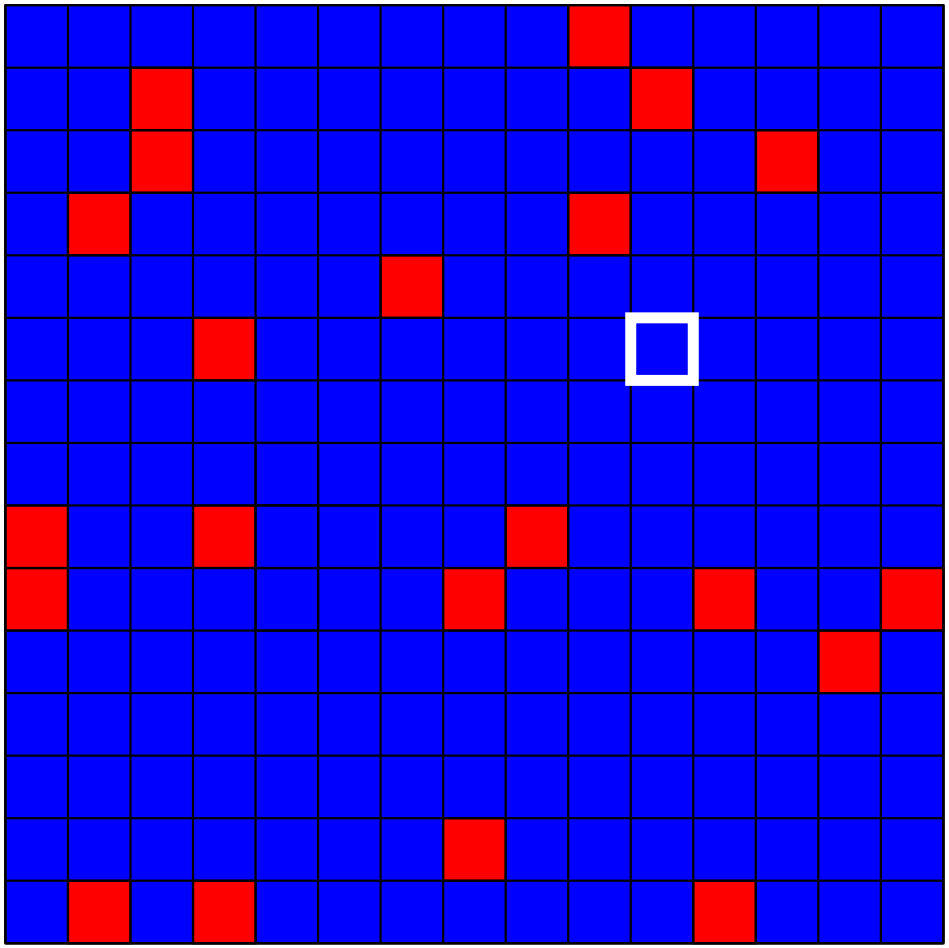


regular lattice with periodic boundary conditions

with prob. q:

- pick random site
- pick random neighbour
- copy feature value from neighbour to site

Model

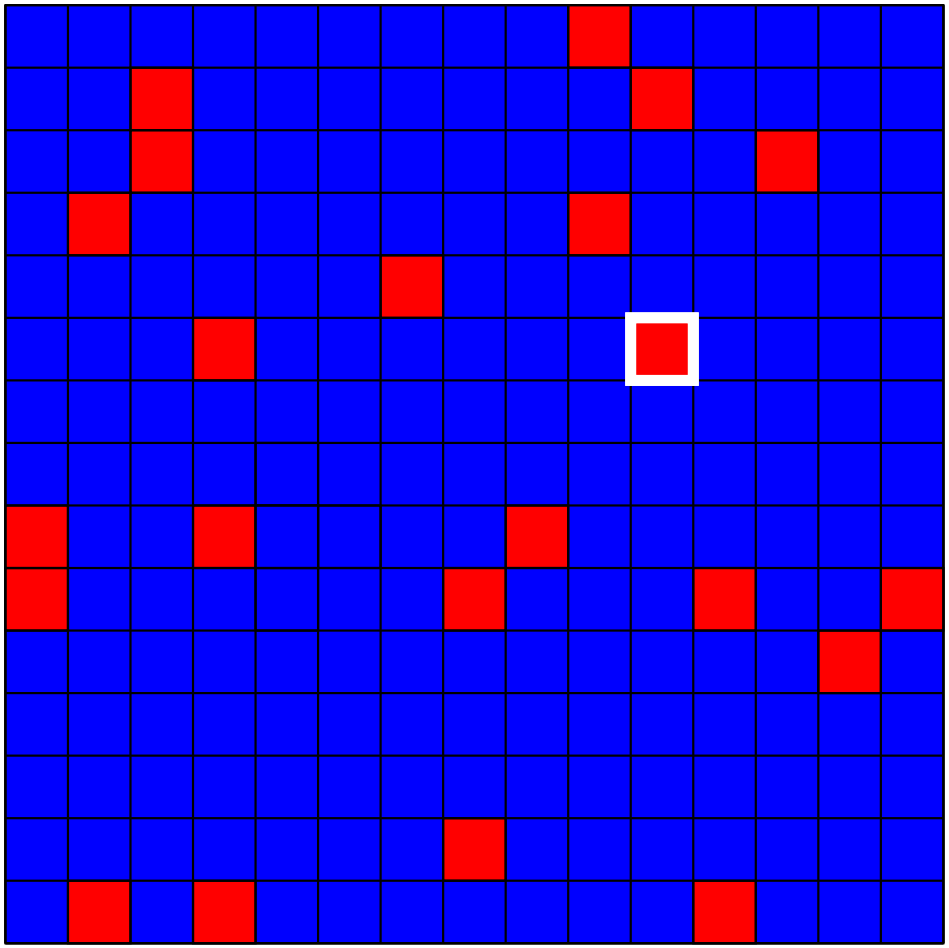


regular lattice with periodic boundary conditions

with prob. $1 - q$:

pick random site

Model



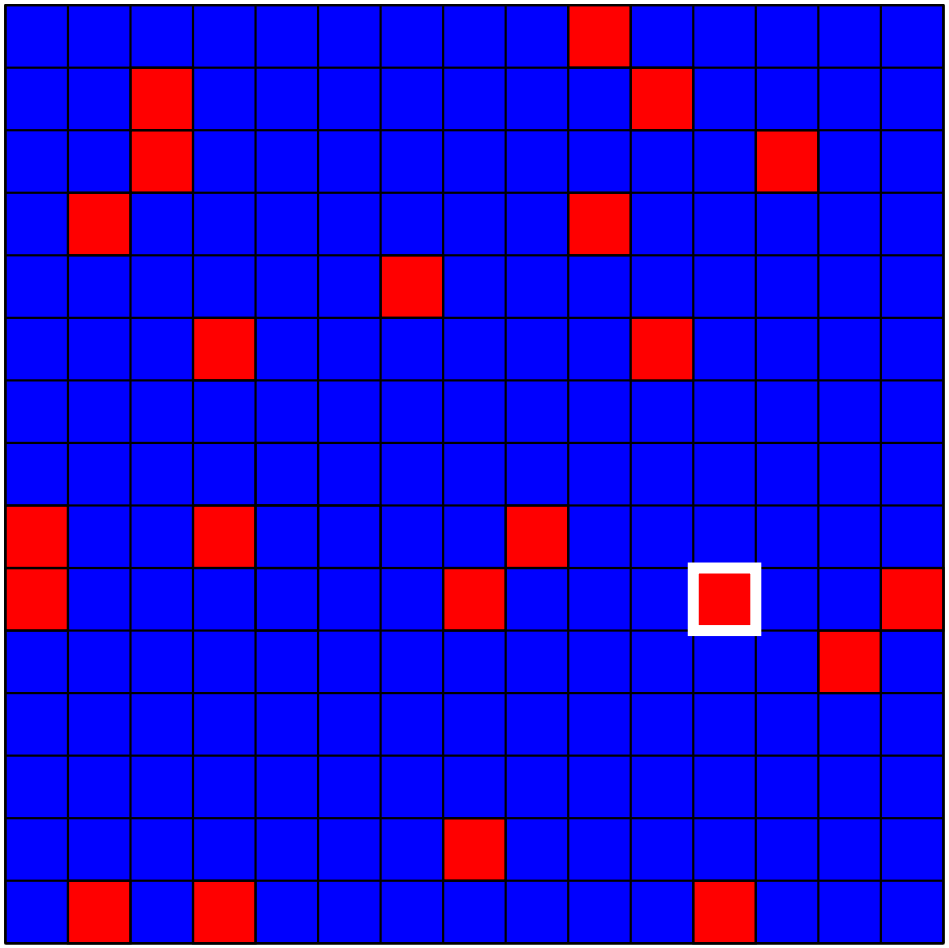
regular lattice with periodic boundary conditions

with prob. $1 - q$:

pick random site

innovate feature with prob. p_i

Model



regular lattice with periodic boundary conditions

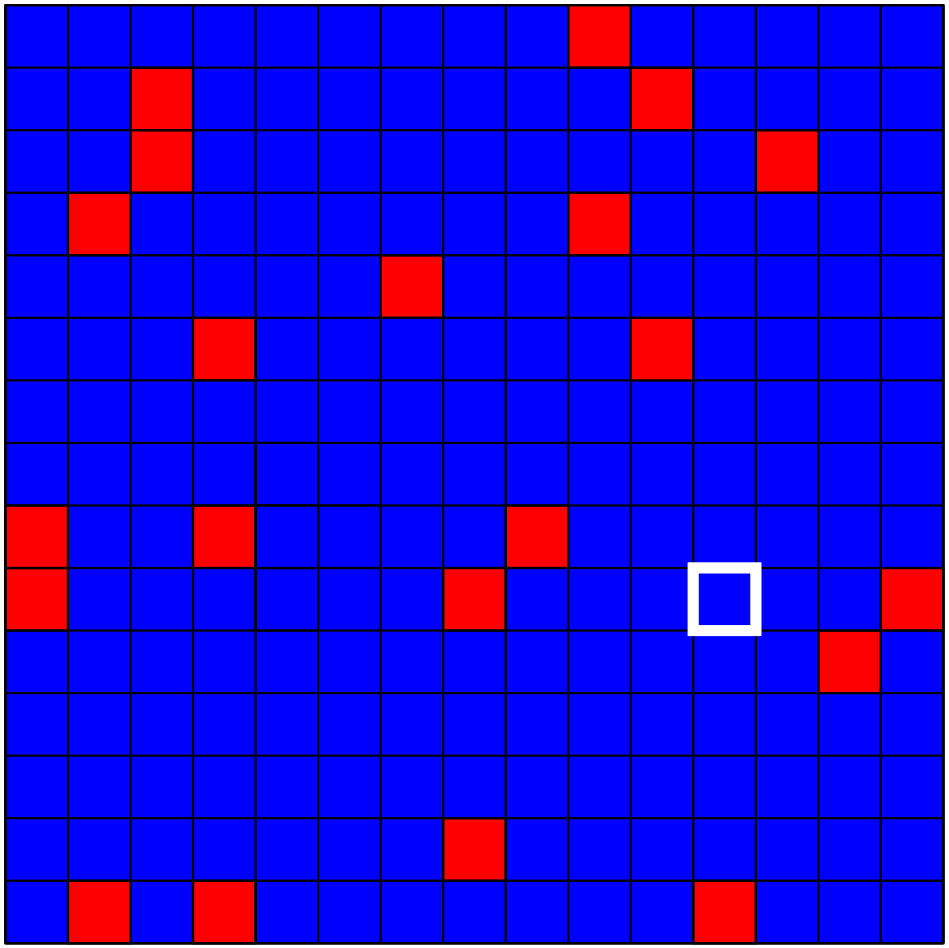
with prob. $1 - q$:

pick random site

innovate feature with prob. p_i

OR

Model



regular lattice with periodic boundary conditions

with prob. $1 - q$:

pick random site

innovate feature with prob. p_i

OR

lose feature with prob. p_e

Dynamics

- Interested in the long-time behaviour of this system
- In principle, could run computer simulations for various parameter combinations and wait
- However, model turns out to be soluble!
- More precisely, the values of ρ and σ at the stationary state can be solved as soon as p_i, p_e and q are known

The stationary state

- Steady-state feature frequency (probability of feature) is

$$\rho = \frac{p_i}{p_i + p_e} = \frac{1}{1 + p_e/p_i}$$

- Steady-state isogloss density is

$$\sigma = h(\tau)\rho(1-\rho)$$

with

$$h(\tau) = \frac{(1+\tau)\pi}{K\left(\frac{1}{1+\tau}\right)} - 2\tau,$$

where $K(\cdot)$ is the complete elliptic integral of the first kind and

$$\tau = \frac{(1-q)(p_i + p_e)}{q}.$$

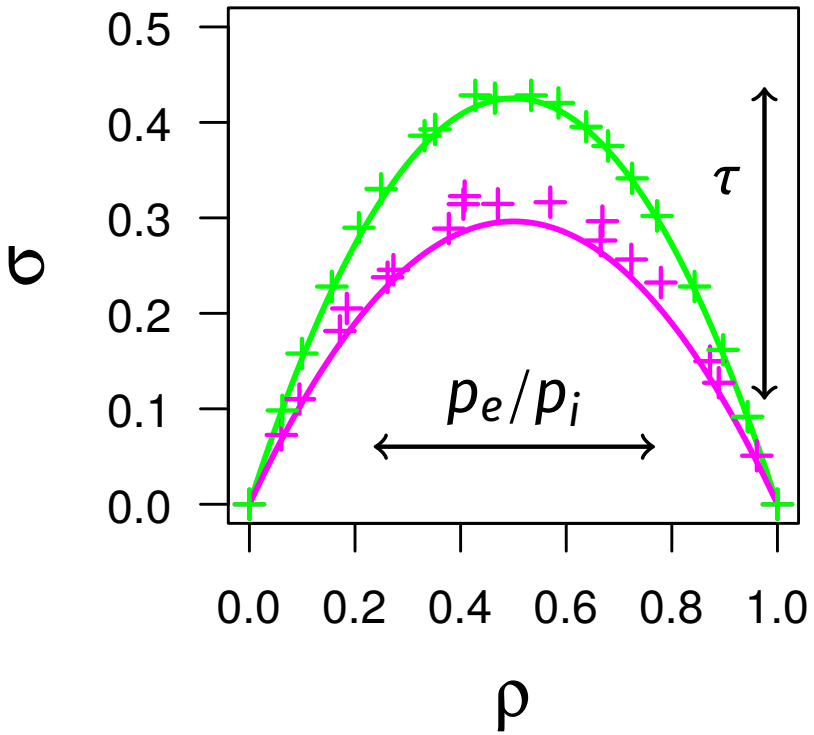
$$\tau = \frac{(1-q)(p_i+p_e)}{q}$$

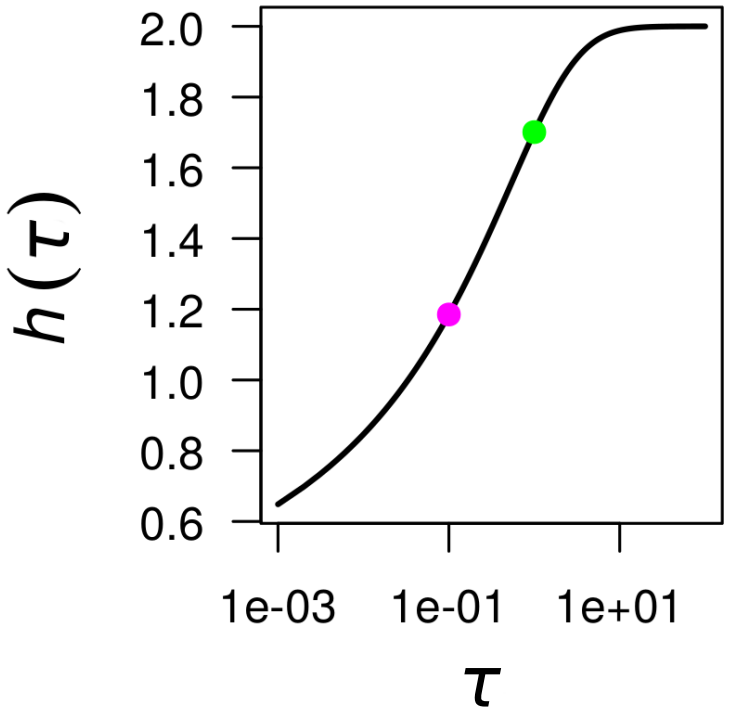
$$\sigma = h(\tau)\rho(1-\rho)$$

Properties of $h(\tau)$:

$h(\tau) \to 0$ as $\tau \to 0$

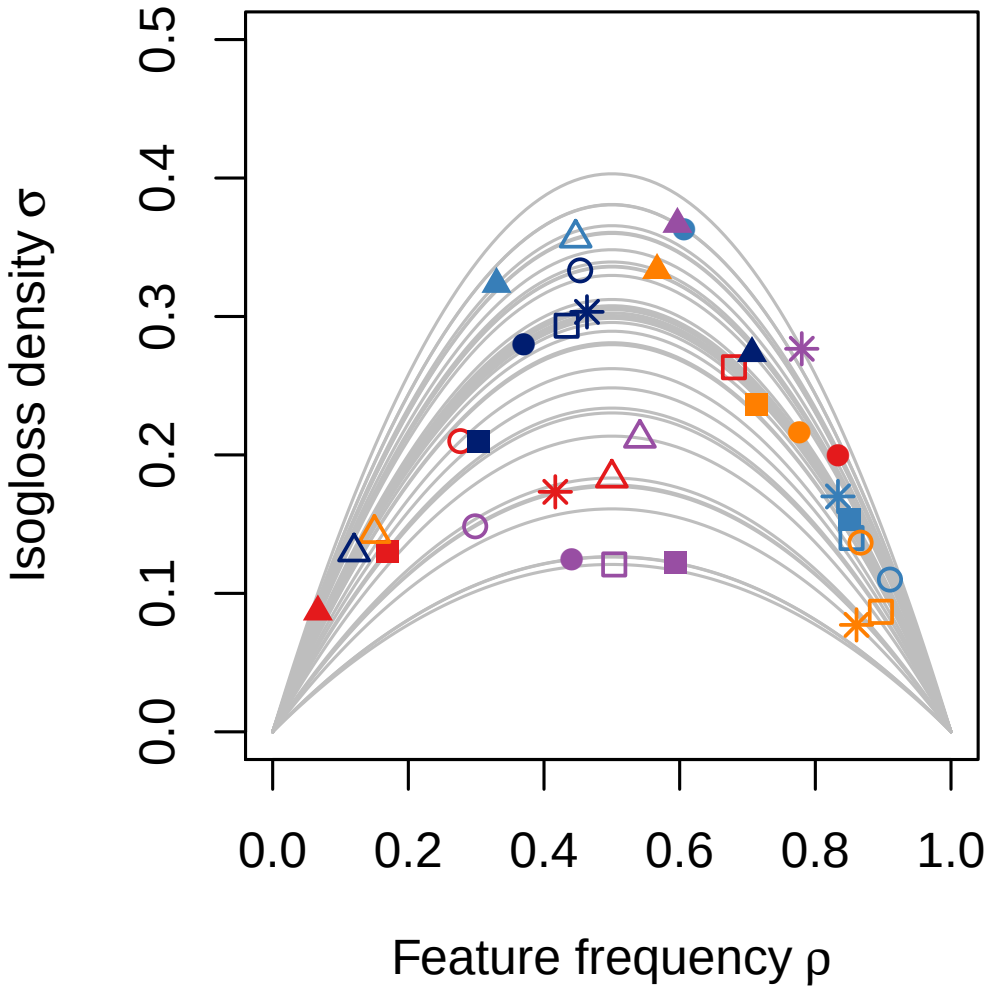
$h(\tau) \to 2$ as $\tau \to \infty$

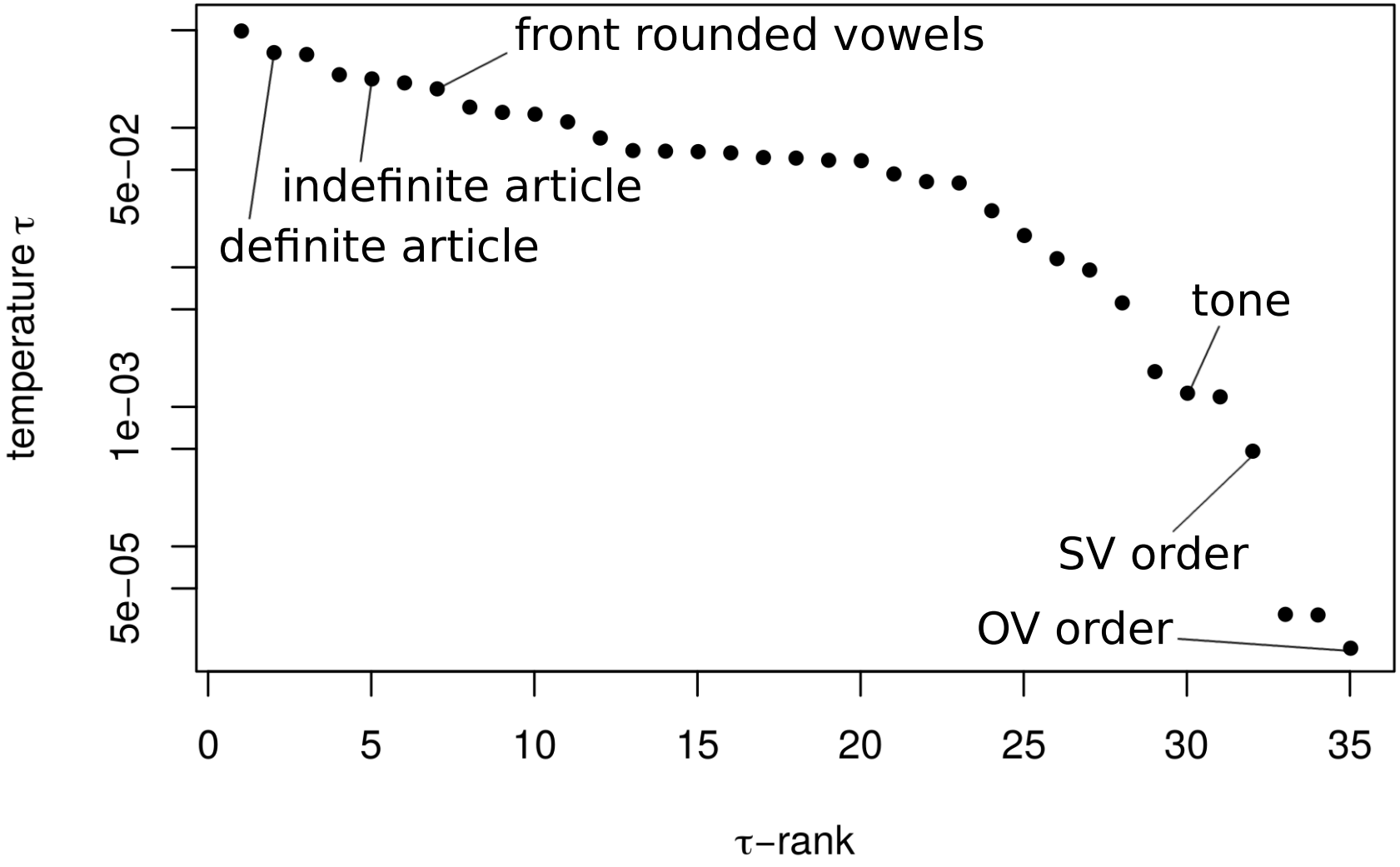




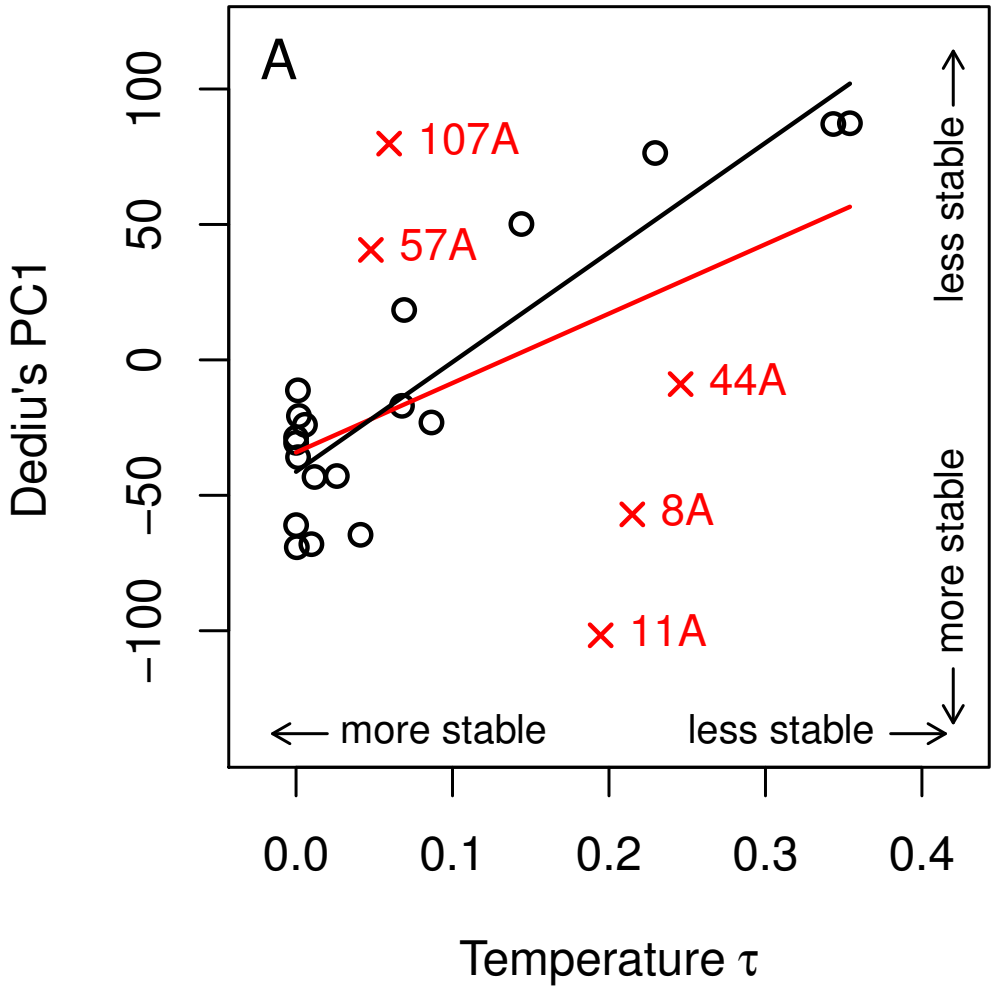
Estimating temperature

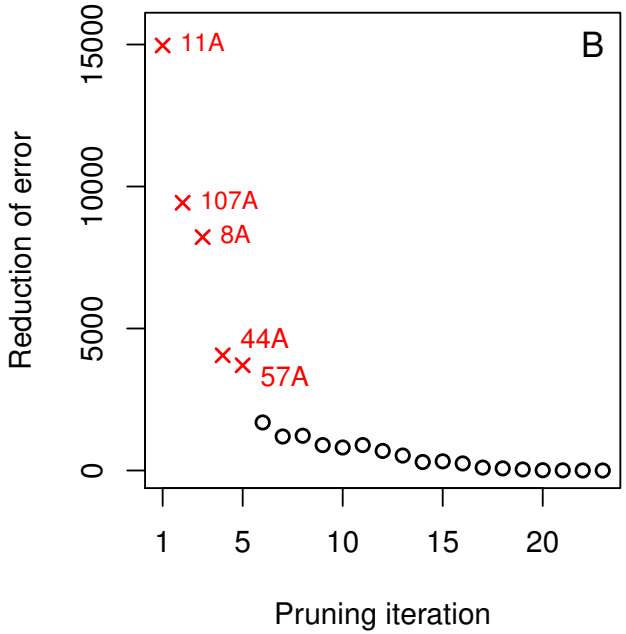
- Each feature lies on a unique parabola in the ρ, σ space
- Solve for $h(\tau)$, invert to recover τ




front rounded vowels
indefinite article
definite article
tone
SV order
OV order
temperature τ
5e−02
1e−03
5e−05
0
5
10
15
20
25
30
35
τ−rank

Comparison with a phylogenetic method[4]





11A = front rounded vowels, 107A = passive construction, 8A = lateral consonants, 44A = gender in independent personal pronouns, 57A = possessive affixes

[4] Dediu (2011)

Front rounded vowels (FRV)

- WALS 11A, the presence of one or more of [y, ø, œ, Œ]
- One of the most stable features under the phylogenetic analysis
- One of the least stable under our analysis
- Evidence seems to be in our favour:
  - FRV are frequently innovated: documented for Armenian, Attic-Ionic Greek, English, French, Frisian, Lithuanian, Old Scandinavian, Oscan, Parachi, Umbrian, West Syriac, Yiddish, Zuberoan Basque, among others (see references in Kauhanen, Gopal, Galla, and Bermúdez-Otero, 2018)
  - FRV can arise through i-mutation (Ohala, 1981)
  - FRV are difficult for L2 learners (Rochet, 1995; Strange, Levy, & Law, 2009), therefore possibly prone to be lost in contact situations
- Similar stories can be told of the remaining outliers (we hope)

Conclusions & future work

Conclusions

- Have investigated empirical spatial distributions of linguistic features
- Have proposed a mathematical model of these and solved its steady-state properties
- Have derived estimates of evolution rates of features without recourse to phylogeny
  - Significant agreement with the phylogenetic method (interesting)
  - Some differences, too (also interesting)

Future work

- Incorporate feature interactions & multivalued features
- Incorporate more realistic spatial (and genetic) dynamics
- Test model against more databases
- Correlate results systematically against previous methods

Thank you!

Preprint

- https://arxiv.org/abs/1801.09637
- http://henr.in/papers

Thanks to

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References I

Bentz, C. & Winter, B. (2013). Languages with more second language learners tend to lose nominal case. *Language Dynamics and Change*, *3*, 1–27.

Castellano, C., Fortunato, S., & Loreto, V. (2009). Statistical physics of social dynamics. *Reviews of Modern Physics*, *81*, 591–646.

Dediu, D. (2011). A Bayesian phylogenetic approach to estimating the stability of linguistic features and the genetic biasing of tone. *Proceedings of The Royal Society B*, *278*, 474–479.

Dediu, D. & Cysouw, M. (2013). Some structural aspects of language are more stable than others: A comparison of seven methods. *PLOS One*, *8*, e55009.

Dryer, M. S. (1989). Large linguistic areas and language sampling. *Studies in Language*, *13*, 257–292.

Dryer, M. S. & Haspelmath, M. (Eds.). (2013). *WALS Online*. Leipzig: Max Planck Institute for Evolutionary Anthropology. Retrieved July 18, 2017, from http://wals.info/

Gardani, F., Arkadiev, P., & Amiridze, N. (Eds.). (2014). *Borrowed morphology*. Berlin: De Gruyter.

Greenberg, J. H. (1978). Diachrony, synchrony, and language universals. In J. H. Greenberg (Ed.), *Universals of human language* (Vol. 1, pp. 61–91). Stanford, CA: Stanford University Press.

Greenhill, S. J., Atkinson, Q. D., Meade, A., & Gray, R. D. (2010). The shape and tempo of language evolution. *Proceedings of the Royal Society B*, *277*, 2443–2450.

References II

Greenhill, S. J., Wu, C.-H., Hua, X., Dunn, M., Levinson, S. C., & Gray, R. D. (2017). Evolutionary dynamics of language systems. *Proceedings of the National Academy of Sciences*, *114*(42), E8822–E8829.

Kauhanen, H., Gopal, D., Galla, T., & Bermúdez-Otero, R. (2018). Geospatial distributions reflect rates of evolution of features of language. *arXiv*. Retrieved from https://arxiv.org/abs/1801.09637

Maslova, E. (2004). Dinamika tipologičeskix raspredelenij i stabil'nost' jazykovyx tipov. *Voprosy jazykoznanija*, *5*, 3–16.

Ohala, J. J. (1981). The listener as a source of sound change. In C. S. Masek, R. A. Hendrick, & M. F. Miller (Eds.), *Papers from the parasession on language and behavior* (pp. 178–203). Chicago Linguistic Society 17. Chicago, IL: Chicago Linguistics Society.

Rochet, B. L. (1995). Perception and production of second-language speech sounds by adults. In W. Strange (Ed.), *Speech perception and linguistic experience: Issues in cross-language research* (pp. 379–410). Baltimore, MD: York Press.

Strange, W., Levy, E. S., & Law, F. F., II. (2009). Cross-language categorization of French and German vowels by naïve American listeners. *The Journal of the Acoustical Society of America*, *126*, 1461–1476.

Wichmann, S. & Holman, E. W. (2009). Temporal stability of linguistic typological features. Manuscript.